

Intermediate Elective 2

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Set-up

Read-in packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# customization
library(prismatic)
library(stringr)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# arranging plots
library(patchwork)

## 1. LOAD PACKAGES & SETUP ----
suppressPackageStartupMessages({
  if (!require("pacman")) install.packages("pacman")
  pacman::p_load(
    tidyverse, # Easily Install and Load the 'Tidyverse'
    ggtext,    # Improved Text Rendering Support for 'ggplot2'
```

```

    showtext,    # Using Fonts More Easily in R Graphs
    janitor,     # Simple Tools for Examining and Cleaning Dirty Data
    scales,      # Scale Functions for Visualization
    glue,        # Interpreted String Literals
    gghighlight # Highlight Lines and Points in 'ggplot2'
  )))

### figure size ----
camcorder::gg_record(
  dir      = here::here("temp_plots"),
  device   = "png",
  width    = 16,
  height   = 20,
  units    = "in",
  dpi      = 320
)

```

Read-in data

```

# WEEK 4 data

veg <- read_csv(here("data", "veg.csv"))

metadata <- read_csv(here("data", "vp_veg_metadata.csv"))

# WEEK 5 data
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
  ↪ Data-2026-03-10.xlsx"),

```

```
sheet = "Water Quality",  
na = c("", "n/a", "N/A", "over", "173+"))
```

Class code

```
# WEEK 4  
  
# create a new object from veg  
pools_explicit0 <- veg |>  
  # filter all occurrences where transect_name contains VP  
  filter(str_detect(transect_name, "VP") &  
    site == "ncos") |>  
  # complete all possible combinations of year, pool, psoc and distance  
  complete(year, transect_name, psoc, transect_distance,  
    # fill values of percent cover with 0  
    fill = list(percent_cover = 0)  
  ) |>  
  # group by year, pool, and psoc  
  group_by(year, transect_name, psoc) |>  
  # calculate total percent cover across all quads  
  summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>  
  # ungroup the data frame  
  ungroup() |>  
  # create a new column called year_pool from year and transect name  
  unite("year_pool", year, transect_name, remove = TRUE) |>  
  # join with metadata data frame  
  left_join(metadata, by = "year_pool") |>  
  # calculate mean percent cover from total percent cover/number of quadrats  
  mutate(mean_pc = sum_pc/num_quad)  
  
# WEEK 5  
  
# creating new object from sites  
ncos_sites <- sites |>  
  # filter to only include NCOS sites sampled four times a year  
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")  
  
# creating new clean object from taxon_list  
taxon_list_clean <- taxon_list |>  
  # converting taxon name to snake case (no spaces or capital letters,
```

```

# only underscores)
mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

# creating new clean object from water quality
water_quality_clean <- water_quality |>
# clean column names
clean_names() |>
# filtering join: only include sites in ncos_sites
semi_join(ncos_sites, by = "site") |>
# create new column to join with later data frames
unite("date_site", date, site, remove = TRUE) |>
# group by date_site column
group_by(date_site) |>
# calculate median pH, salinity, DO
summarize(
  med_ph = median(p_h, na.rm = TRUE),
  med_sal = median(salinity_ppt, na.rm = TRUE),
  med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(
  date, sample_type, site,
  ostracod,
  copepod,
  hemiptera_corixidae_boatman,
  cladocera,
  nematode,
  diptera_ceratopogonidae,
  annelida_oligochaete,
  diptera_chironomid,
  annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>

```

```

# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
  ↪ liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↪ = TRUE),
       site_full = fct_rev(site_full))

```

1. Explain your inspiration

In 1-2 sentences each, describe:

The visualization I chose for my inspiration was [Steven Ponce's "Evolution of MTA Art Materials \(1980-2020\)"](#). This makes sense for the data sets that I am utilizing because I am trying to find the relationship between vegetation and its explanatory variables as well as invertebrate abundance and its explanatory variables. The circular graph will be able to show

the abundance of both vegetation and invertebrates, as well as a statistical summary of the mean.

- which variables from your dataset will be shown in your visualization, and what visual components they map onto (axes, shapes, colors, etc.)

2. Plan your figure

[Visualization plan](#) (please see README)

3. Code your figure

Vegetation

Vegetation graphs code

```
# create new object from veg
native_spp <- veg |>
# filter only native observations
  filter(cover_category == "NATIVE COVER") |>
# pull to extract data frame column as a vector
  pull(psoc)

# random sample
sample(native_spp, 10)
```

```
[1] "Epilobium_brachycarpum"    "Psilocarphus_brevissimus"
[3] "Juncus_bufonius"          "Distichlis_spicata"
[5] "Extriplex_californica"    "Erigeron_canadensis"
[7] "Salicornia_pacifica"      "Cyperus_eragrostis"
[9] "Grindelia_camporum"       "Stipa_pulchra"
```

```
# create new object from pools_explicit0
species_rich <- pools_explicit0 |>
# group by year and pool
  group_by(year, transect_name) |>
# filter data frame with vector \nto get native species,
```

```

# also filter percent cover to identify species presence
filter(psoc %in% native_spp, mean_pc > 0) |>
# species richness calculation
summarize(spp_rich = n_distinct(psoc)) |>
mutate(transect_name = case_when(
  transect_name == "VP-01" ~ "Pool 1",
  transect_name == "VP-02" ~ "Pool 2",
  transect_name == "VP-03" ~ "Pool 3",
  transect_name == "VP-04" ~ "Pool 4",
  transect_name == "VP-05" ~ "Pool 5",
  transect_name == "VP-06" ~ "Pool 6",
  transect_name == "VP-07" ~ "Pool 7",
  transect_name == "VP-08" ~ "Pool 8"
))
# # ungroup data frame
# ungroup()

# sample 10 rows
slice_sample(species_rich, n = 10)

```

```

# A tibble: 40 x 3
# Groups:   year [5]
   year transect_name spp_rich
  <dbl> <chr>         <int>
1  2021 Pool 7         17
2  2021 Pool 6         13
3  2021 Pool 8         19
4  2021 Pool 1         13
5  2021 Pool 3         19
6  2021 Pool 5         12
7  2021 Pool 4         15
8  2021 Pool 2         14
9  2022 Pool 5          9
10 2022 Pool 1         17
# i 30 more rows

```

```

species_rich_final <- species_rich |>
  group_by(year) |>
  mutate(mean_pool = mean(spp_rich)) |>
  ungroup()

```

```
# species_rich_final$year <- as.factor(species_rich_final$year)

species_rich_final$year <- rep(2021:2025, each = 8)
```

Vegetation visualization

```
colors <- scale_color_manual(
  values = c(
    "Pool 1" = "#285943",
    "Pool 2" = "#D36582",
    "Pool 3" = "#82204A",
    "Pool 4" = "#74546A",
    "Pool 5" = "#C490D1",
    "Pool 6" = "#FFB7C3",
    "Pool 7" = "#5FB0B7",
    "Pool 8" = "#6C91BF"
  ))

species_rich_final$year <- as.numeric(species_rich_final$year)
```

```
p <- ggplot(data = species_rich_final,
  aes(x = year,
    y = spp_rich,
    colour = transect_name,
    fill = transect_name
  )) +

  # Geoms
  geom_line(size = 1.7,
    alpha = 0.9) +
  gghighlight(
    use_direct_label = FALSE,
    unhighlighted_params = list(
      linewidth = 0.7,
      alpha = 0.75,
      color = "gray60"
    )
  ) +
  # Scales
  colors +
```



```

scale_y_continuous(
  expand = c(0, 0),
  limits = c(0, 30)
) +
# Labs
labs(
  title = "MAIN TITLE",
  subtitle = "Species richness over time (2021-2025)\nGrouped by vernal
    ↪ pool",
  x = "Year",
  y = "Species richness (n)"
) +
# Facets
facet_wrap(~transect_name,
  scales = "free", ncol = 2
) +
# Theme
theme(
  plot.title = element_text(
    face = "bold",
    lineheight = 1.1,
    margin = margin(t = 5, b = 5)
  ),
  plot.subtitle = element_text(
    lineheight = 1.2,
    margin = margin(t = 5, b = 20)
  ),
  plot.caption = element_markdown(
    hjust = 0,
    margin = margin(t = 15, b = 5, l = 0),
    lineheight = 1.3
  ),
  strip.text = element_text(
    size = 12,
    face = "bold",
    color = "gray20",
    margin = margin(t = 5, b = 5)
  ),
  strip.background = element_rect(
    fill = "gray95",
    color = "white",
    linewidth = 0.5
  ),

```

```

    panel.spacing = unit(1.5, "lines"),
  ) +
  theme_bw() +
  theme(legend.position = "none")

```

Invertebrates

Invertebrate code

```

# new object from aq_ins_clean
log_aq_ins <- aq_ins_clean |>
  # mutate to get taxon group -species
  mutate(taxon = case_when(
    str_detect(taxon, "annelida") ~ "Annelida",
    str_detect(taxon, "cladocera") ~ "Cladocera",
    str_detect(taxon, "copepod") ~ "Copepod",
    str_detect(taxon, "diptera") ~ "Diptera",
    str_detect(taxon, "hemiptera") ~ "Hemiptera",
    str_detect(taxon, "nematode") ~ "Nematode",
    str_detect(taxon, "ostracod") ~ "Ostracod"
  )) |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  subset(!is.na(med_do)) |>
  arrange(desc(log_ave_lit))

# sample of ten rows
slice_sample(log_aq_ins, n = 10)

```

A tibble: 10 x 24

	date_site	date	site	taxon	ave_lit	med_ph	med_sal	med_do
	<chr>	<dtm>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	2023-03-03_NDC	2023-03-03 00:00:00	NDC	Anne~	0	7.45	0.688	5.64
2	2023-05-16_NVBR	2023-05-16 00:00:00	NVBR	Ostr~	0.133	9.29	NA	9.83
3	2020-02-21_NVBR	2020-02-21 00:00:00	NVBR	Anne~	0	7.34	4.1	6.1
4	2021-11-16_NPB	2021-11-16 00:00:00	NPB	Cope~	0.533	7.3	2.38	12.1
5	2024-01-28_NPB	2024-01-28 00:00:00	NPB	Dipt~	0	7.43	3.08	6.6
6	2023-05-06_NDC	2023-05-06 00:00:00	NDC	Dipt~	0	10.1	NA	13.1
7	2022-02-21_NPB	2022-02-21 00:00:00	NPB	Cope~	23.5	7.39	34.1	0.34

```

8 2023-05-05_NPB 2023-05-05 00:00:00 NPB Cope~ 10 NA NA 12.1
9 2023-11-12_NMC 2023-11-12 00:00:00 NMC Hemi~ 0.133 NA 47 14.0
10 2023-09-15_NMC 2023-09-15 00:00:00 NMC Anne~ 0 NA NA 14.2
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>

```

Invertebrate visualization

```

# creating colors object for ggplot
colors2 <- scale_fill_manual(
  values = c(
    "Devereux Creek" = "#07beb8",
    "North of Venoco Bridge" = "#3dccc7",
    "South of Phelps Bridge" = "#68d8d6",
    "Phelps Road" = "#9ceae8",
    "South of Venoco Bridge" = "#c4fff9",
    "North of Phelps Bridge" = "#b2f7ef",
    "Main Channel" = "#bee9e8"
  )
)

```

```

p2 <- ggplot(data = log_aq_ins,
  aes(x = taxon,
    y = log_ave_lit,
    fill = site_full
  )) +

# geometries
geom_jitter(height = 0,
  shape = 21,
  width = 0.1) +

# adding median points
stat_summary(geom = "point",
  fun = "median",
  size = 3,
  color = "darkblue") +

gghighlight(
  use_direct_label = FALSE,
  unhighlighted_params = list(
    linewidth = 0.7,
    alpha = 0.75,

```

```

    color = "gray60"
  )
) +
# Scales
colors2 +
scale_y_continuous(
  expand = c(0, 0),
  limits = c(-0.3, 7)
) +
# Labs
labs(
  subtitle = "Median Aquatic invertebrate abundance/L at NCOS sites
    ↪ (2020-2024)",
  caption = "Data derived from ENVS193DD course, UCSB
    Visualization credit Steven Ponce (https://github.com/poncest)",
  x = "Taxon",
  y = "log +1 transformed average abundance/L"
) +
# Facets
facet_wrap(~site_full,
  scales = "free", ncol = 2
) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 10)) +
# Theme
theme(
  plot.title = element_text(
    face = "bold",
    lineheight = 1.1,
    margin = margin(t = 5, b = 5)
  ),
  plot.subtitle = element_text(
    lineheight = 1.2,
    margin = margin(t = 5, b = 20)
  ),
  plot.caption = element_markdown(
    hjust = 0,
    margin = margin(t = 15, b = 5, l = 0),
    lineheight = 1.3
  ),
  strip.text = element_text(
    size = 12,
    face = "bold",

```

```

    color = "gray20",
    margin = margin(t = 5, b = 5)
  ),
  strip.background = element_rect(
    fill = "gray95",
    color = "white",
    linewidth = 0.5
  ),
  panel.spacing = unit(1.5, "lines"),
) +
theme_bw() +
theme(legend.position = "none")

```

Putting it all together

p / p2

4. In 1-3 sentences each, describe:

- your process of using someone else's code to create a visualization
- challenges you encountered as you made your visualization
- how your visualization differs from visualizations you made in class, for assignments, or for a previous elective
- update readme for right packages